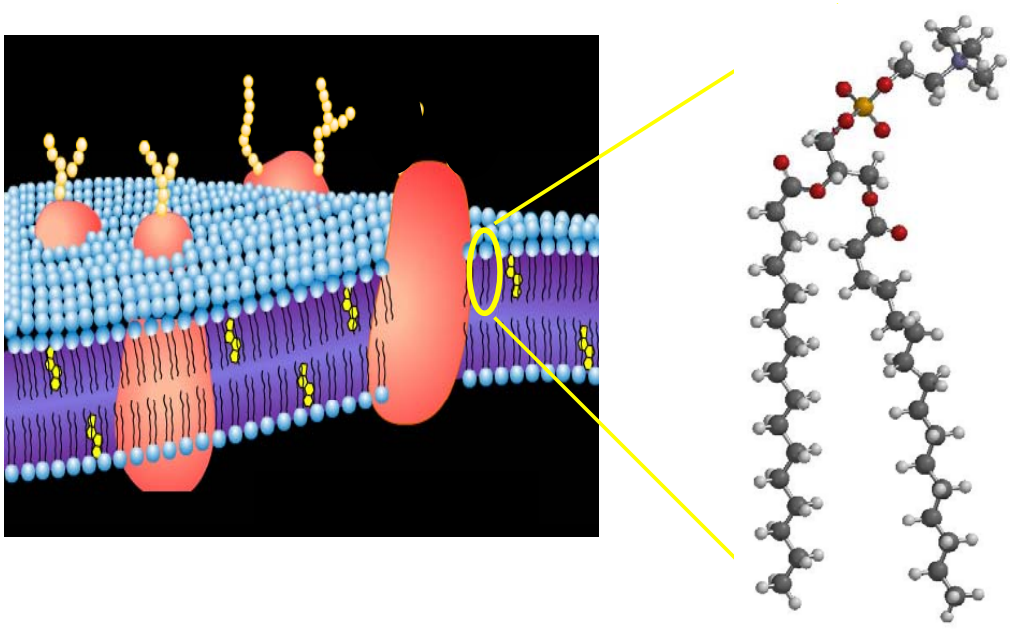
화학특강 (2011년 2학기)
제3장 Molecular Self-Assembly

중요 학습 내용

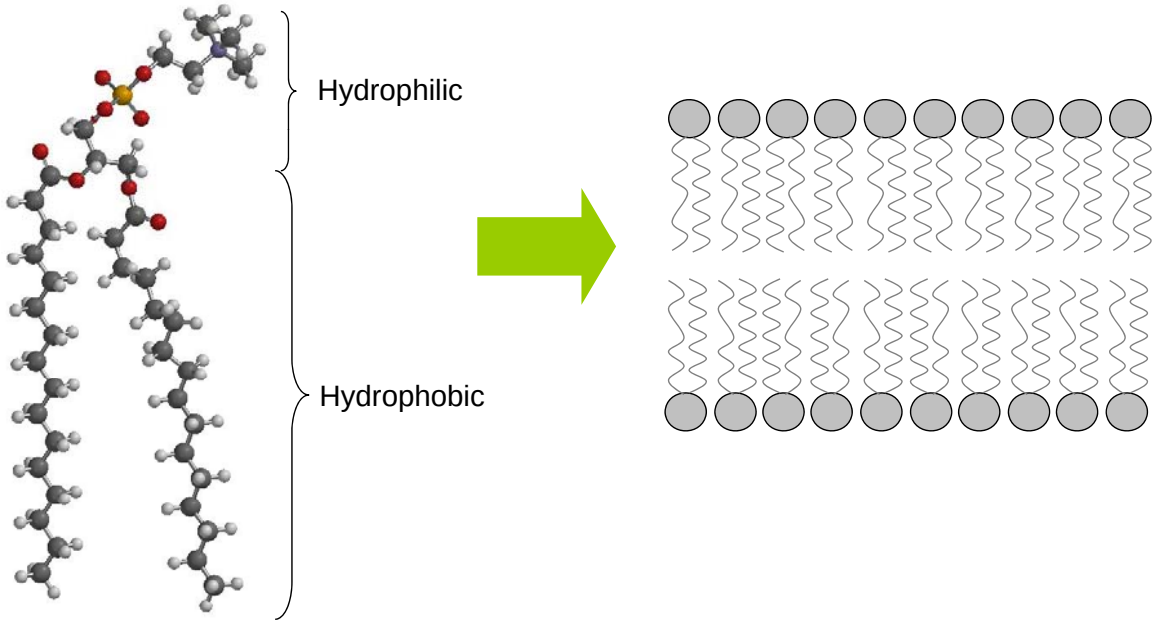
- Self-assembly of lipid molecules
- Principles of self-assembly
- Examples of self-assembly
- Self-assembled monolayers (SAM)
- Colloids and self-assembly

Cell Membranes and Lipid Molecules

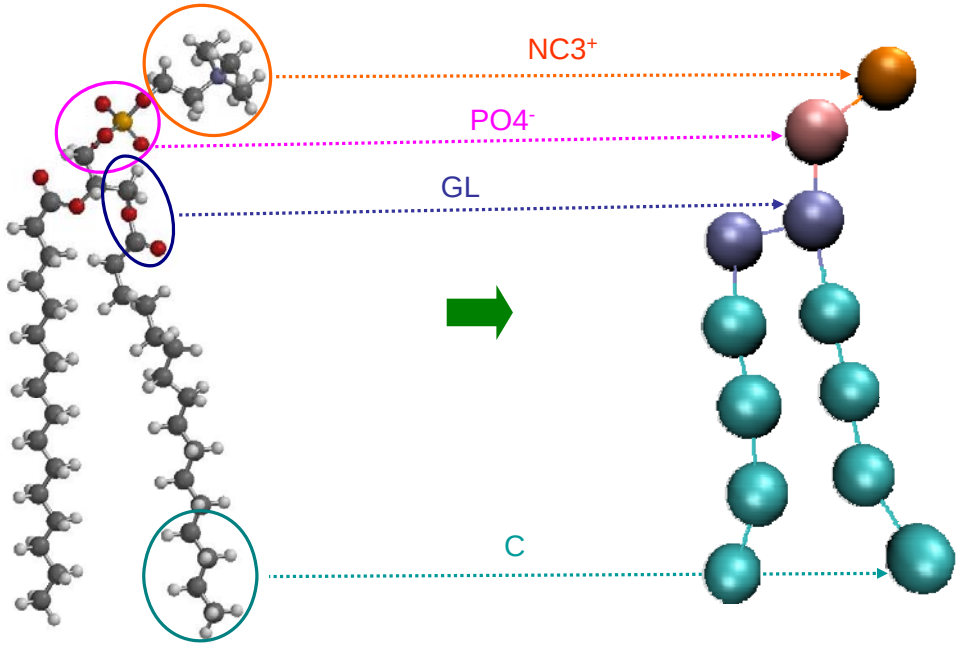


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Lipid molecule as an amphiphile



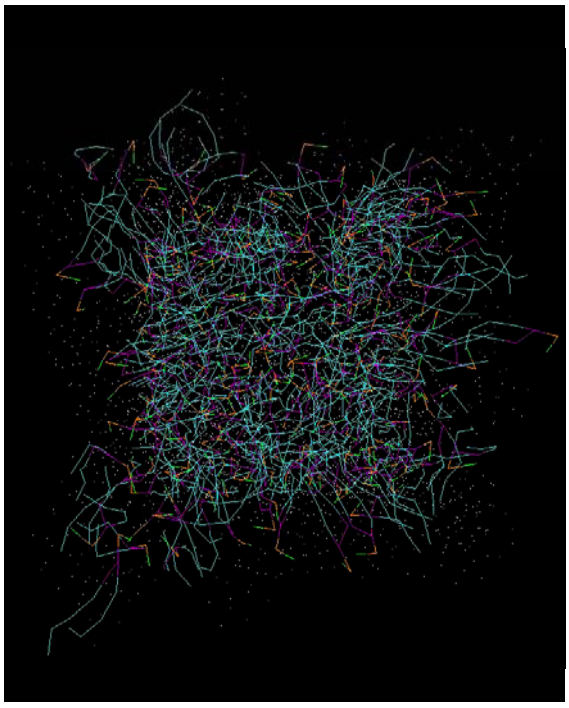
CG modeling for DPPC lipid molecules



all atom model

coarse grained model

Self-Assembly of Lipid Molecules



Principles of Self-Assembly (1)

1. Self-assembly is a process in which components, either separate or linked, spontaneously form ordered aggregates.
2. Self-assembly can occur with components having sizes from the molecule (molecular self-assembly) to the macroscopic, provided that appropriate conditions are met.
3. In molecular self-assembly, the molecular structure determines the structure of the assembly.
4. The sucess of self-assembly in a molecular system is determined by five characteristics of the system.

   1) Components: A self-assembling system consists of a group of molecules or segments of macromolecule that interact with one another.

Principles of Self-Assembly (2)

2) **Interactions**: Self-assembly occurs when molecules interact with one another through a balance of attractive and repulsive interactions. This interactions are generally weak and noncovalent (van der Waals, Coulomb interactions, hydrophobic interactions, and hydrogen bonds) interactions.

3) **Reversibility (equilibrium):** For self-assembly to generate ordered structures, the association either must be reversible or must allow the components to adjust their positions within an aggregate once it has formed.

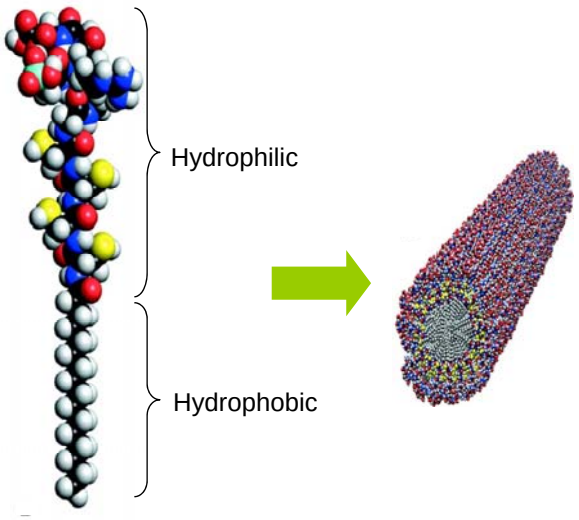
4) **Environment**: The self-assembly of molecules normally is carried out in solution or at an interface to allow the required motion of the components.

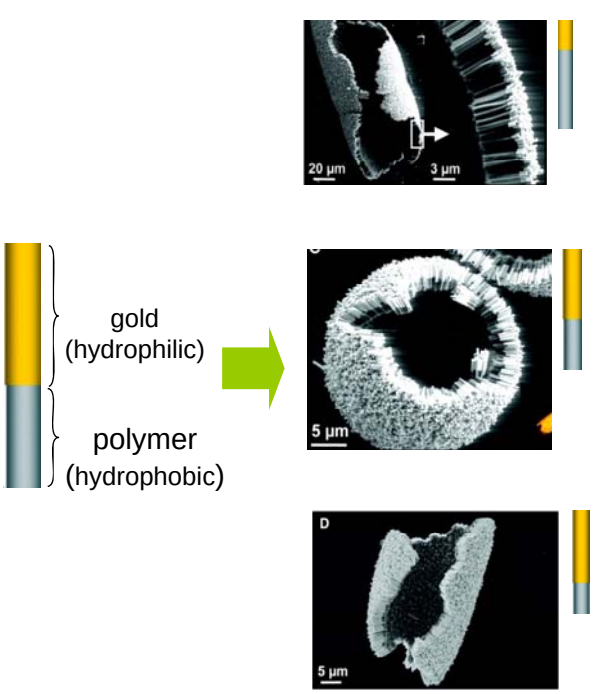
5) **Mass transport and agitation**: For self-assembly to occur, the molecules must be mobile. In solution, thermal motion provides the major part of the motion required to bring the molecules into contact.

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Principles of Self-Assembly

- Components

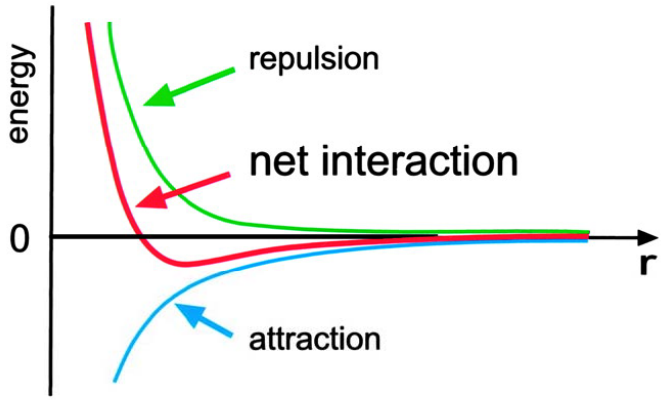


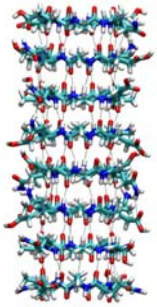


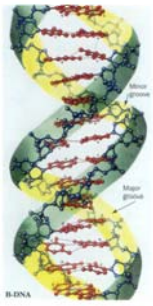
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Principles of Self-Assembly

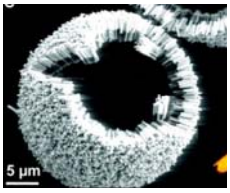
- Interactions







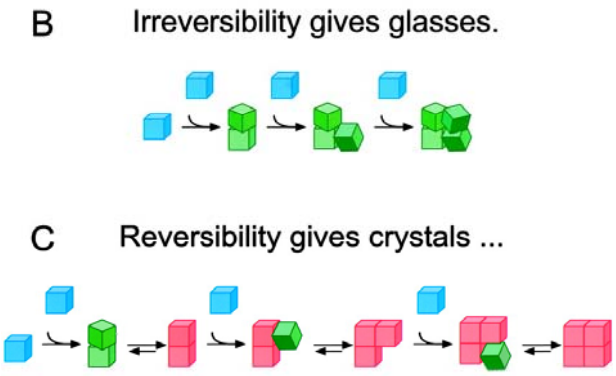




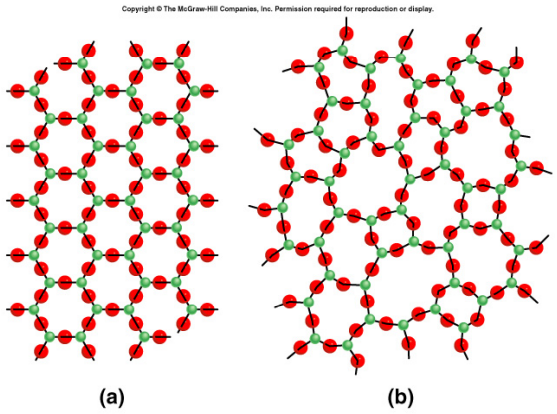
G.M Whitesdies and M. Boncheva, Proc. Nat. Acad. Sci. USA **98**, 4835 (2002)

Principles of Self-Assembly

- Reversibility (equilibrium)



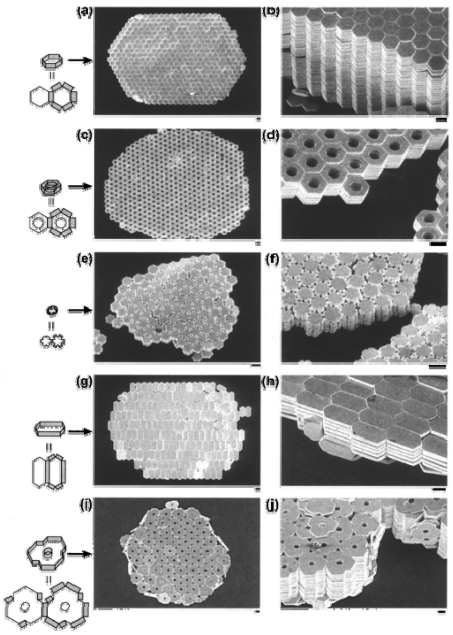
Crystalline quartz (a) and noncrystalline quartz (b)



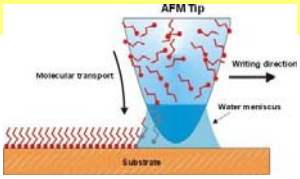
Chang, 11.7

Principles of Self-Assembly

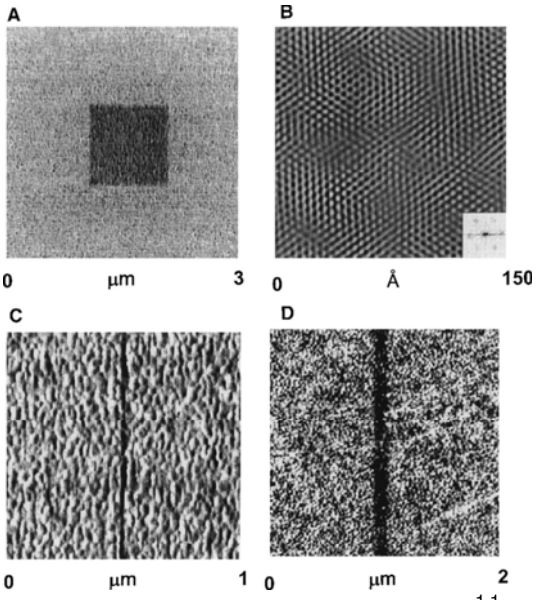
- Environment



Whitesdies et al., JACS **123**, 7677 (2001)

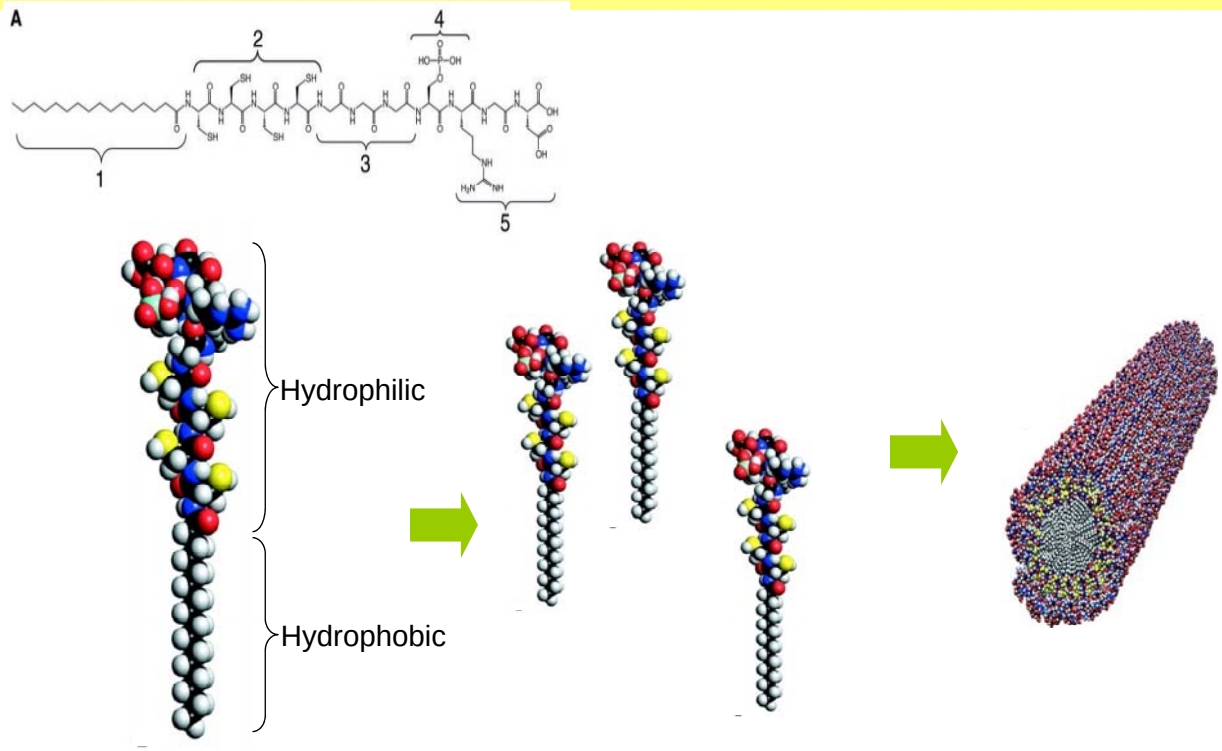


DPN on gold and on mica



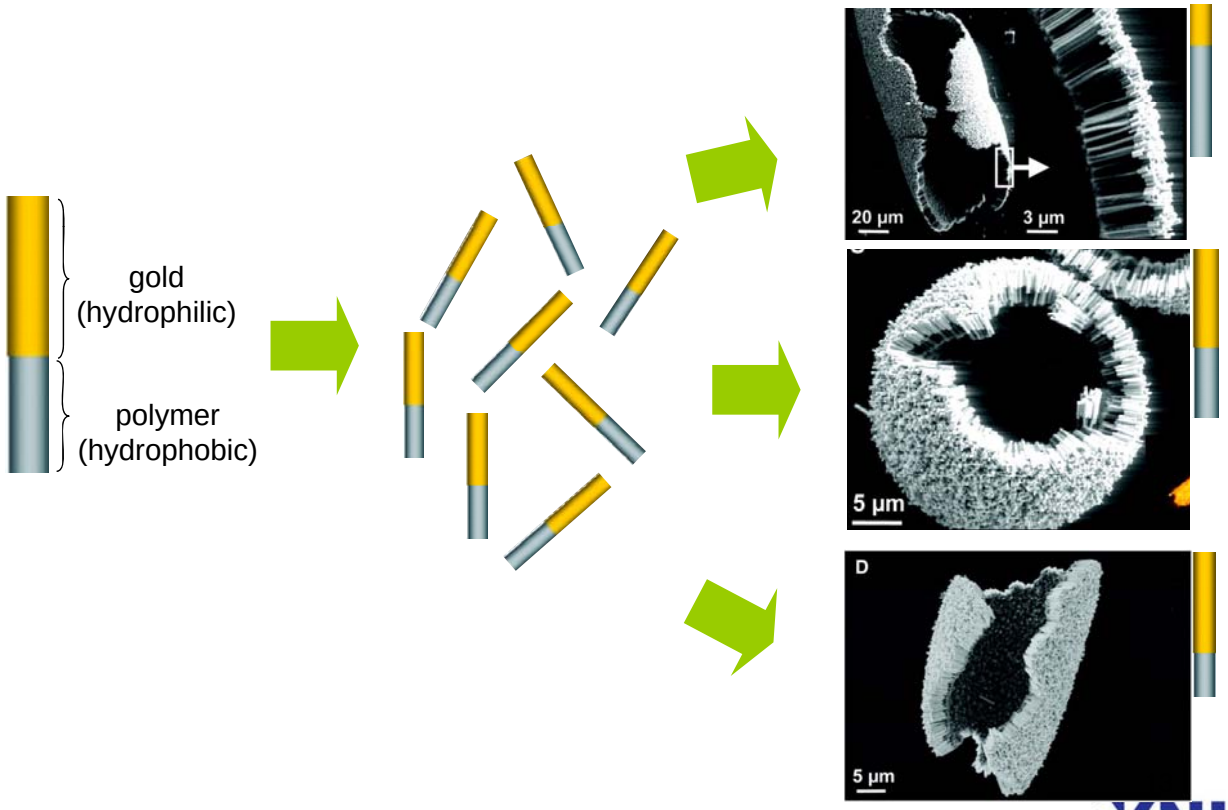
Mirkin et al., Science **283**, 661 (1999)

Self-Assembly of Peptide Amphiphile Nanofibers



Hartgerink et al., Science **294**, 1684 (2001)

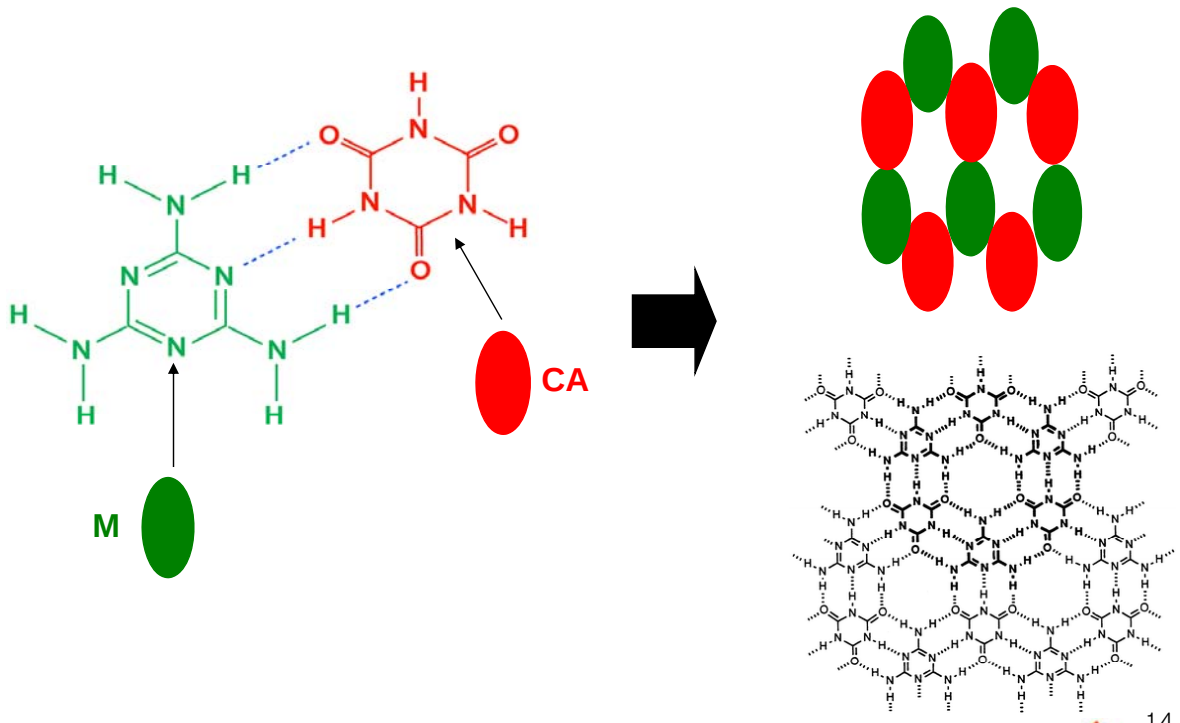
Self-assembly of metal-polymer amphiphiles



Park et al., Science **303**, 348 (2004)

Molecular Self-Assembly

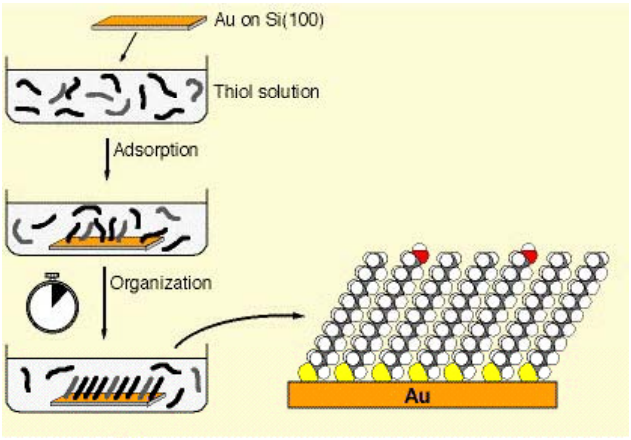
- Melamine (M) - cyanuric acid (CA) complex (hydrogen bonds)

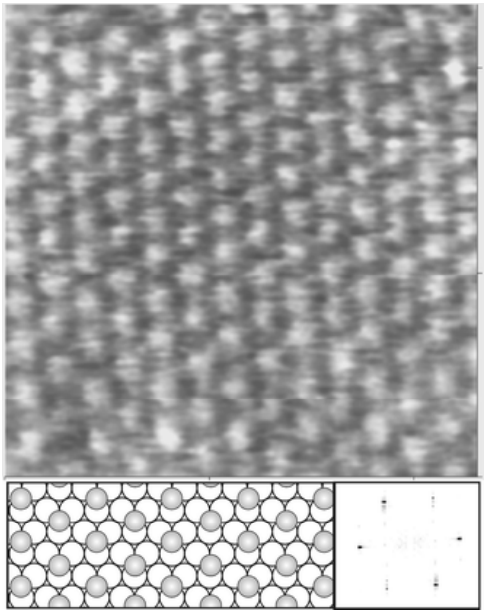


Seto and Whitesdies, JACS **112**, 6409 (1990)

Self-Assembled Monolayers (SAM)

- Alkanethiolates on Au



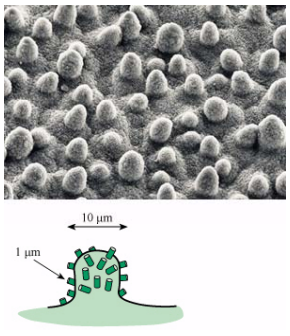


Hexanethiolates on Au[111]
Vericat et al., PCCP **7**, 3258 (2005)

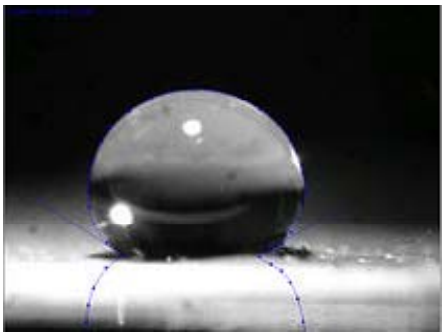
Alkanethiolates are tilted approximately 30° from the normal to the surface to maximize the van der Waals interactions between adjacent methylene group

Self-Cleaning Surface

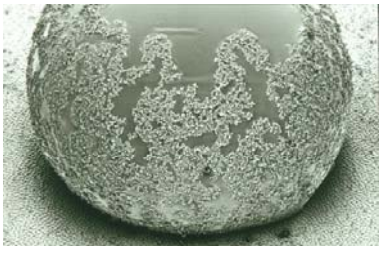
- Lotus effect: the very high water repellency (superhydrophobicity) exhibited by the leaves of the lotus flower.



연잎 표면



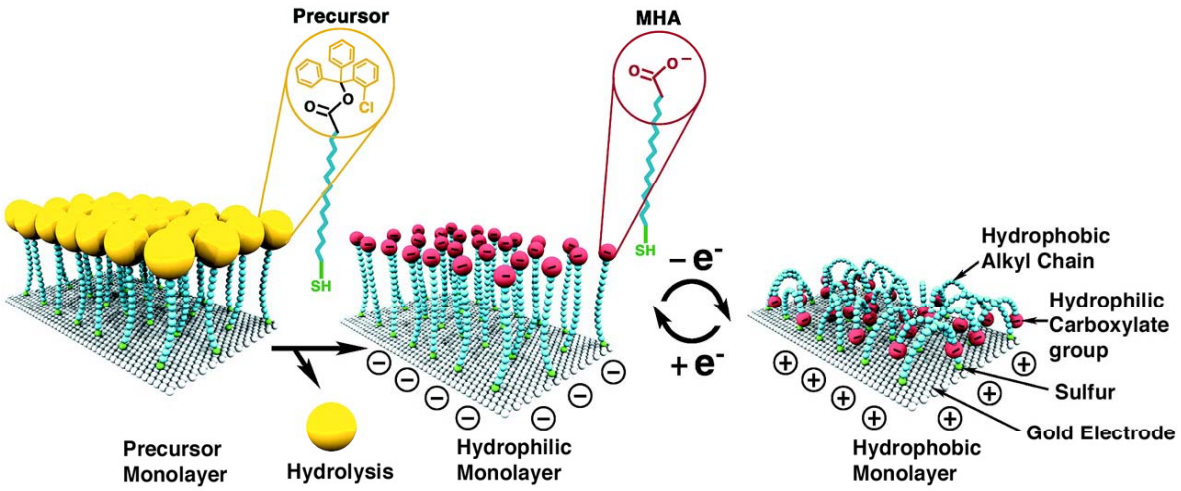
연잎 표면 위의 물방울



연잎 표면 위의 물방울에
붙어 있는 먼지 등

SAM Applications: Electrowettability Switch

- Surface engineering: hydrophilic monolayer ↔ hydrophobic monolayer

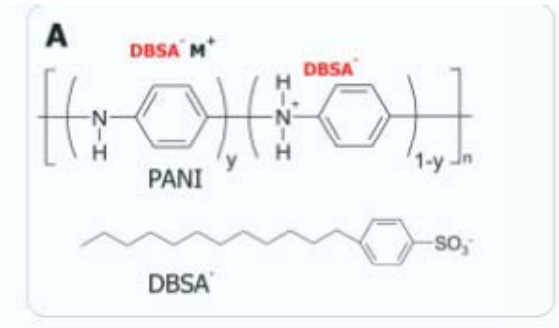


MHA = (16-mercapto)hexadecanoic acid

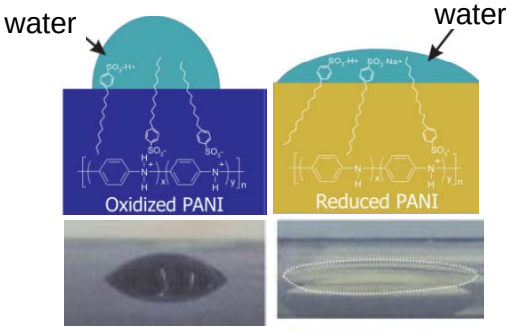
Lahann et al., Science **299**, 371 (2003)

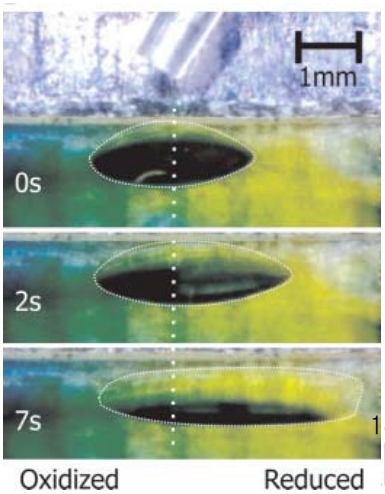
SAM Applications: Electrowettability Switch

- Surface engineering: the combination of a redox-active conducting polymer and a surfactant



PANI: polyaniline (surfactant)
DBSA: dodecylbenzenesulfonic acid (conducting polymer)

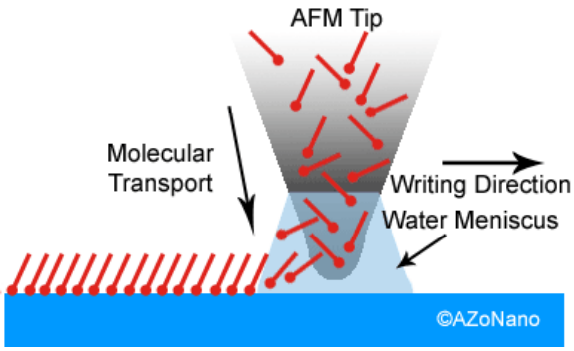


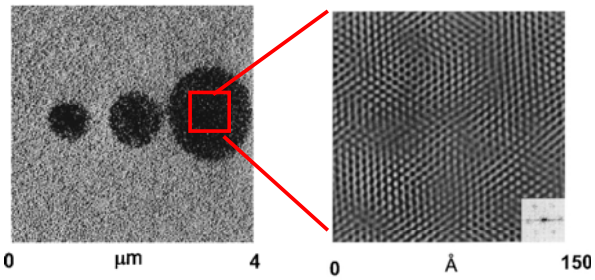


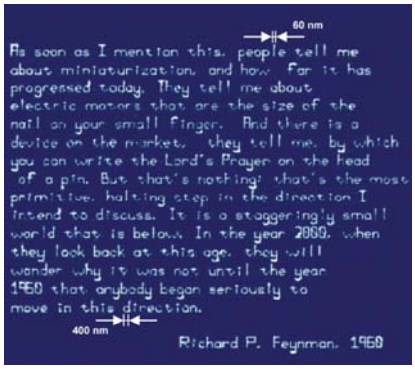
Isaksson et al., Adv. Mater. **16**, 316 (2004)

Patterning SAM via Dip Pen Nanolithography (DPN)

- Coated AFM tip is scanned across a substrate, depositing chemical “ink” in its wake.





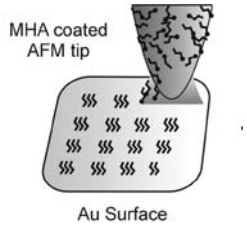


Mirkin et al., Science **283**, 661 (1999)

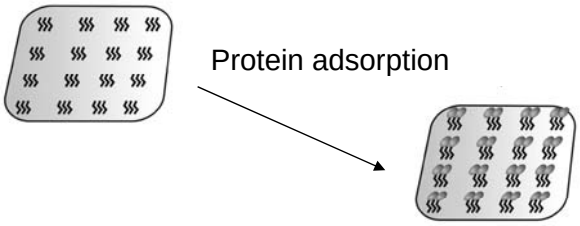
SAM Applications

- SAM as a template for biomolecules

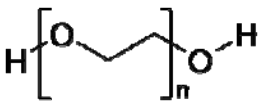
16-hexadecanoic acid
(16-mercaptohexadecanoic acid, MHA)



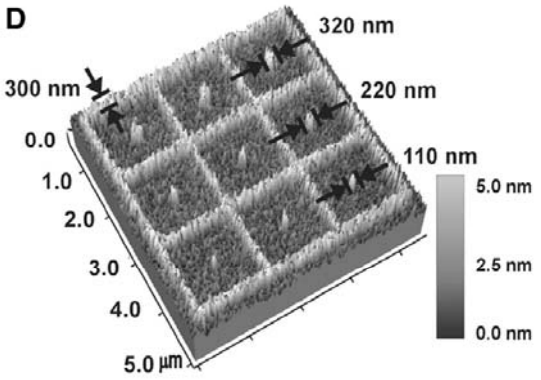
Passivated with PEG-SH



Protein adsorption



Polyethylene glycol (PEG)

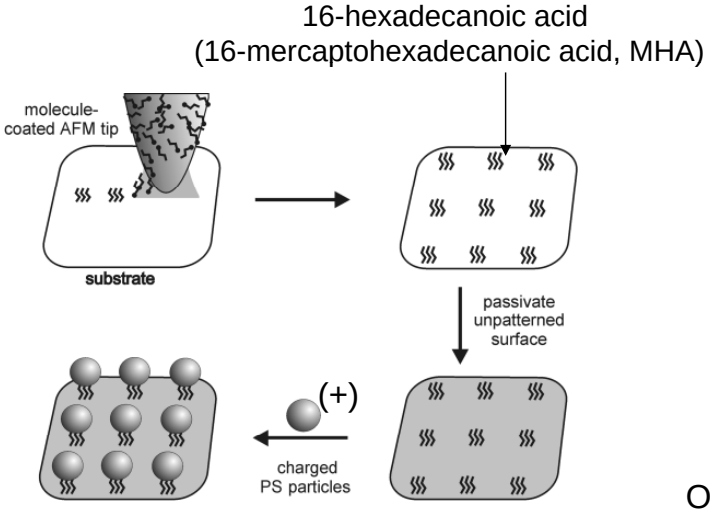


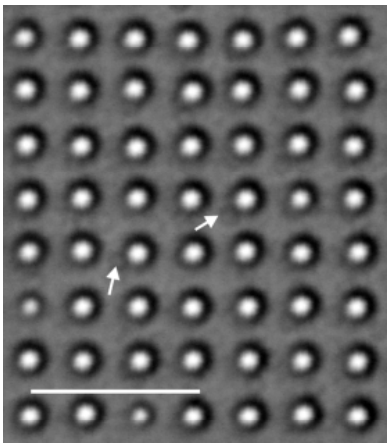
Lysozyme nanoarray

Mirkin et al., Science **295**, 1702 (2002)

SAM Applications

- SAM as a templates: electrostatic assembly of charged polystyrene particles





Optical micrograph of 1.0 μm amine-modified PS particles on MHA templates

Mirkin et al., Angew. Chem. Int. Ed. **40**, 3069 (2001)

Colloids (1)

- Colloids: a dispersion of particles of one substance (the dispersed phase) throughout a dispersing medium maid of another substance

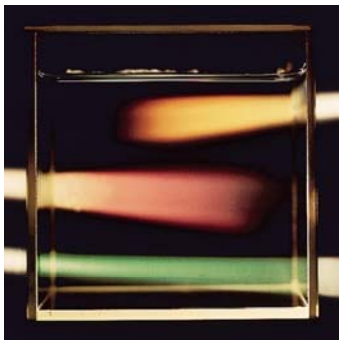
TABLE 12.4 Types of Colloids

Dispersing Medium	Dispersed Phase	Name	Example
Gas	Liquid	Aerosol	Fog, mist
Gas	Solid	Aerosol	Smoke
Liquid	Gas	Foam	Whipped cream
Liquid	Liquid	Emulsion	Mayonnaise
Liquid	Solid	Sol	Milk of magnesia
Solid	Gas	Foam	Plastic foams
Solid	Liquid	Gel	Jelly, butter
Solid	Solid	Solid sol	Certain alloys (steel), gemstones (glass with dispersed metal)

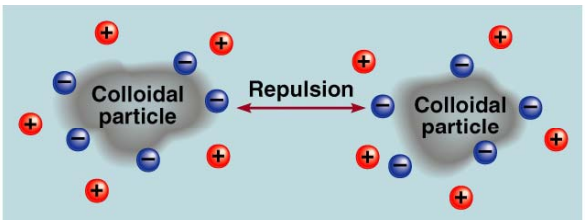
Chang, General Chemistry **12.8**

Colloids (2)

- Tyndall effect



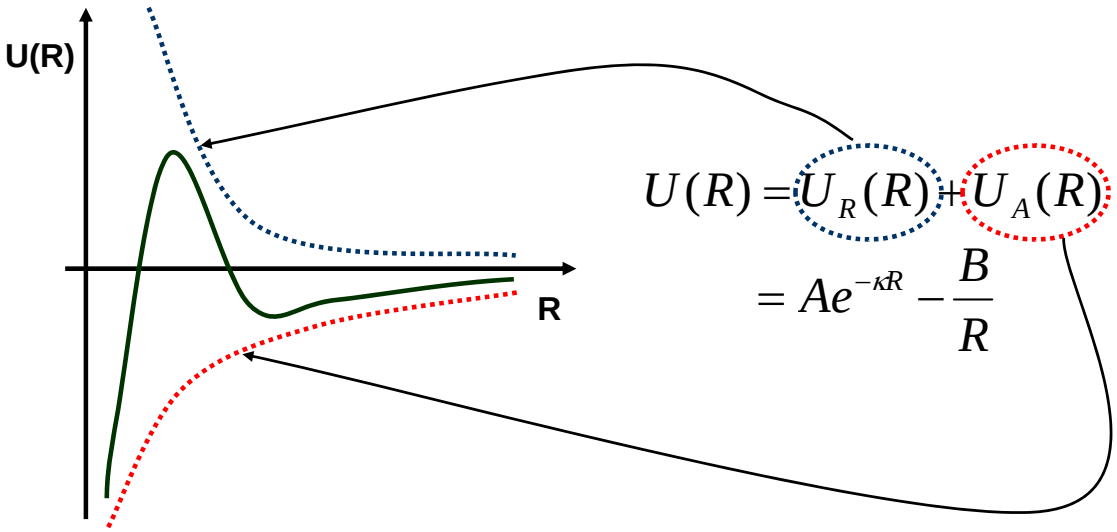
- Electrostatic repulsion



Chang, General Chemistry **12.8**

Colloids (3)

- Interactions between colloids



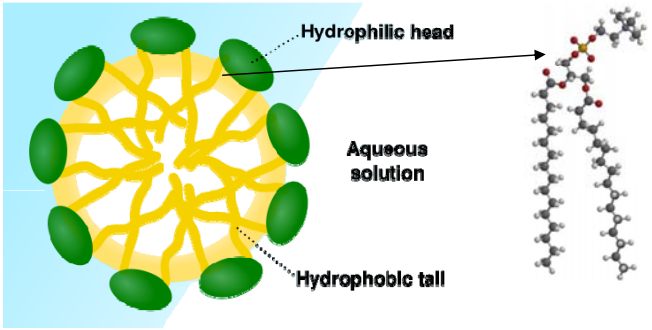
- Coagulation: the destruction of a colloid

Chang, General Chemistry **12.8**

Colloids (4)

- Mayonnaise





- River Delta



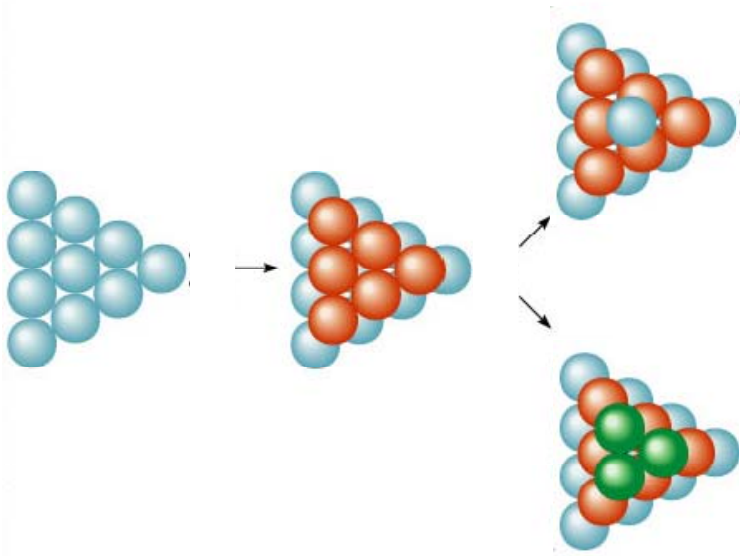
Nile Delta

Chang, General Chemistry **12.8**

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Colloids and Crystrallization

- Hexagonal and cubic closed packed structures



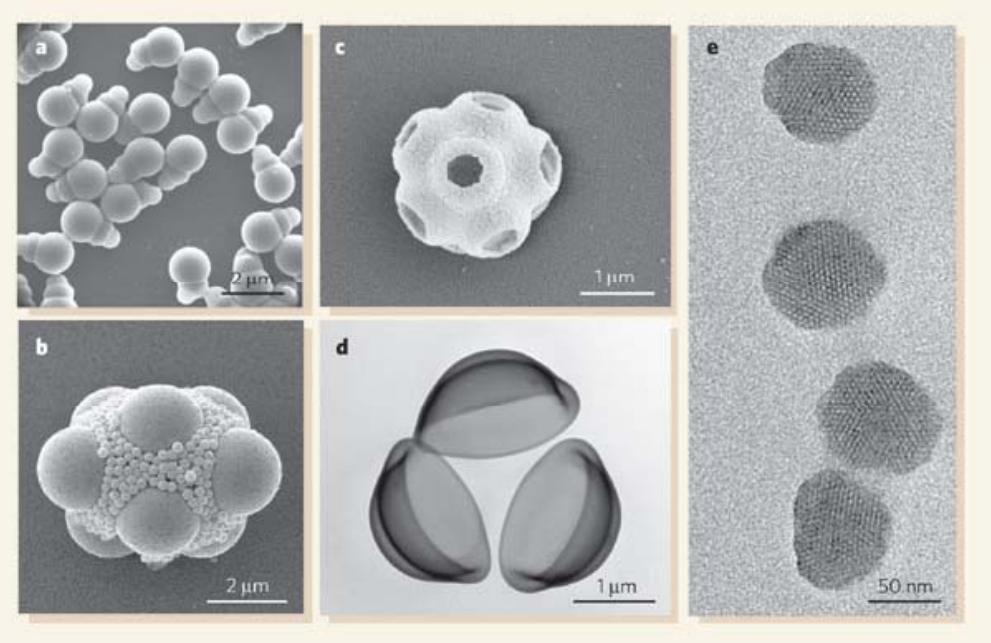
hexagonal closed packed structure (aba)

cubic closed packed structure (abc)

Chang, General Chemistry **11.**

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Colloids in the Nanotechnology

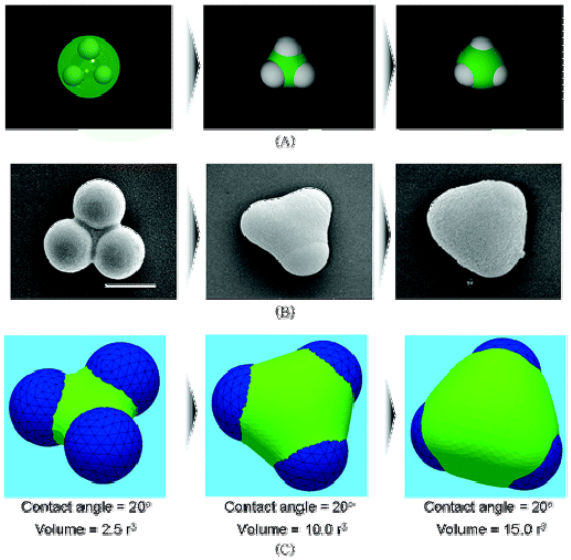


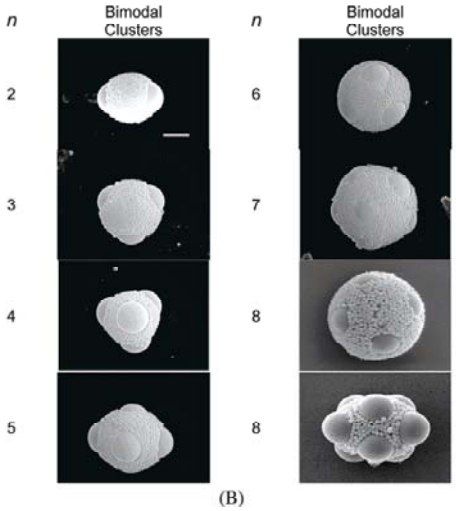
van Blaaderen, Nature **439**, 545 (2006)

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Colloids in the Nanotechnology

- Formation of a trimer cluster of silica microparticles with silica nanoparticles from water-in-oil emulsions



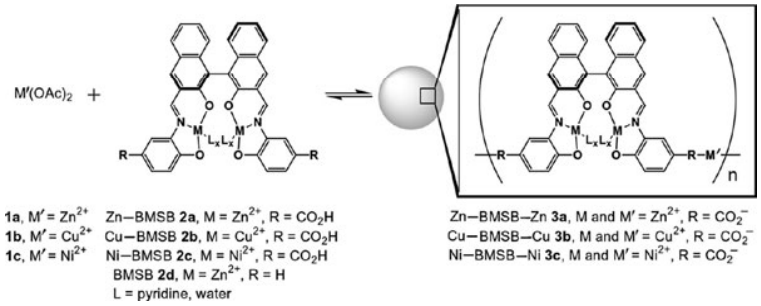


Young-Sang Cho, J. Am. Chem. Soc. **127**, 15968 (2005)

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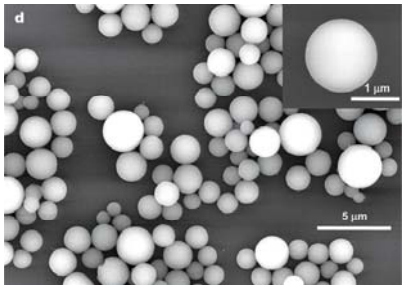
Colloids in the Nanotechnology

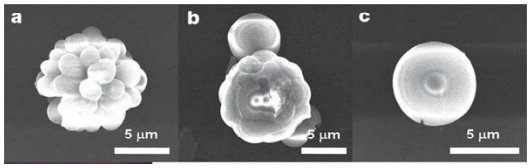
- assembly of metal ions and homochiral carboxylate-functionalized binapthyl bis-metallo-tridentate Schiff base (BMSB) building blocks

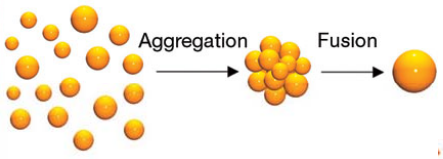


1a, $M' = Zn^{2+}$ Zn–BMSB **2a**, $M = Zn^{2+}$, $R = CO_2H$
1b, $M' = Cu^{2+}$ Cu–BMSB **2b**, $M = Cu^{2+}$, $R = CO_2H$
1c, $M' = Ni^{2+}$ Ni–BMSB **2c**, $M = Ni^{2+}$, $R = CO_2H$
BMSB **2d**, $M = Zn^{2+}$, $R = H$
L = pyridine, water

Zn–BMSB–Zn **3a**, M and $M' = Zn^{2+}$, $R = CO_2^-$
Cu–BMSB–Cu **3b**, M and $M' = Cu^{2+}$, $R = CO_2^-$
Ni–BMSB–Ni **3c**, M and $M' = Ni^{2+}$, $R = CO_2^-$







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